Assignment 2

Q1. Write a C program that reads an input file containing an iterative algorithm (only for loop-based, no recursion) and computes its time complexity by counting the total number of iterations executed.

Specification:

Input: A text file (e.g., algo.txt) whose first line contains the value of n, followed by the algorithm's loop structure written as loop headers with nesting indicated by braces.

The programme must:

Read the file and parse each for loop header — its initial value, terminating condition, and update expression.

Maintain nesting depth using brace matching.

Simulate the loops to count the exact number of times the innermost statement executes for the given n.

Derive the growth function from the update type:

increment/decrement by a constant → linear

multiply/divide by a constant → logarithmic

bound dependent on outer loop variable → dependent (triangular)

Report the exact iteration count and the corresponding asymptotic complexity.

Sample Input File():

```
n = 8
for (i = 1; i <= n; i++) {
   for (j = i; j <= n; j++) {
      count++;
   }
}
```

Expected Output:

Value of n : 8

Loop 1 (depth 1) : i = 1 to n, i++ -> linear

Loop 2 (depth 2) : j = i to n, j++ -> dependent

Total iterations : 36

Growth function : n(n+1)/2

Time Complexity : O(n^2)

Test Cases to be demonstrated:

Loop Structure in File	Iterations (n = 8)	Growth Function	Complexity
`for(i=1;i<=n;i++)`	8	n	O(n)
`for(i=1;i<=n;i++){ for(j=1;j<=n;j++) }`	64	n^2	$O(n^2)$
`for(i=1;i<=n;i=i*2)`	3	$\log_2 n$	O(log n)
`for(i=1;i<=n;i++){ for(j=1;j<=n;j=j*2) }`	24	$n \cdot \log_2 n$	O(n log n)
`for(i=1;i<=n;i++){ for(j=i;j<=n;j++) }`	36	n(n+1)/2	$O(n^2)$

Constraints:

Only for loops are to be handled; while, do-while, and recursive calls need not be processed.

Loop variables are single characters; bounds are either constants, n, or an outer loop variable.

Maximum nesting depth may be assumed to be 3.

Q2. Write a C program to implement Linear Search on an array.

Q3. Write a C program to implement Binary Search on a sorted array.

Q4. Write a C program to implement Selection Sort.

Q5. Write a C program to implement Bubble Sort.